

# MPS-300 Series Modbus Communication Protocol

Register definitions:

| Register address | Name        | 03 Function Code | 06 Function Code | 10 Function Code | Instructions                                                                           |
|------------------|-------------|------------------|------------------|------------------|----------------------------------------------------------------------------------------|
| 0X0000           | Remote Mode | Y                | Y                | Y                | Remote control status 0: Local mode 1: Remote mode                                     |
| 0X0001           | V_SET       | Y                | N                | Y                | Voltage setting register float type                                                    |
| 0X0003           | A_SET       | Y                | N                | Y                | Current setting register float type                                                    |
| 0X0005           | OVP_SET     | Y                | N                | Y                | Overvoltage parameter setting register float type                                      |
| 0X0007           | OCP_SET     | Y                | N                | Y                | Overcurrent parameter setting register float type                                      |
| 0X0009           | OVP_STATE   | Y                | Y                | Y                | Overvoltage status setting register 0: Protection disabled 1: Protection enabled       |
| 0X000A           | OCP_STATE   | Y                | Y                | Y                | Overcurrent status setting register 0: Protection disabled 1: Protection enabled       |
| 0X000B           | OUTPUT      | Y                | Y                | Y                | Output status setting register 0: Output disabled 1: Output enabled                    |
| 0X000C           | STATE       | Y                | Y                | Y                | Status read clear register u16 type, refer to the status register bit definition table |
| 0X000D           | V_OUT       | Y                | N                | N                | Output actual voltage register float type                                              |
| 0X000F           | A_OUT       | Y                | N                | N                | Output actual current register float type                                              |
| 0X0011           | CV/CC       | Y                | N                | N                | Output operating status register 0: CV mode 1: CC mode                                 |

Status Register (0x000C) Bit Definition Table:

| Bit | Name     | Properties | Instructions                                                                                                             |
|-----|----------|------------|--------------------------------------------------------------------------------------------------------------------------|
| 2   | OTP_FLAG | W/R        | 1: Overtemperature protection flag. Write 1 to clear the overtemperature flag (when temperature returns to a safe value) |
| 1   | OCP_FLAG | W/R        | 1: Overcurrent protection flag. Write 1 to clear the overcurrent flag                                                    |
| 0   | OVP_FLAG | W/R        | 1: Overvoltage protection flag. Write 1 to clear the overvoltage flag                                                    |

**Note: All float-type parameters occupy two register addresses**

Function Code Table and Interval Time are as follows:

| Function Code | Corresponding function                                 | Interval time between two operations |
|---------------|--------------------------------------------------------|--------------------------------------|
| 0X03          | Command to read one or multiple consecutive registers  | N * 5ms                              |
| 0X06          | Command to write a single register                     | 10ms                                 |
| 0X10          | Command to write one or multiple consecutive registers | N * 5ms                              |

Note: N is the number of registers



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The communication protocol format is as follows:

## Function Code 0X03

PC sends: 8 Byte

| Address | Function code | Starting address high byte | Starting address low byte | Number of registers high byte | Number of registers low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|----------------------------|---------------------------|-------------------------------|------------------------------|-------------------------|--------------------------|
| 0X01    | 0X03          |                            |                           |                               |                              |                         |                          |

Power supply returns: 5+N\*2 Byte

| Address | Function code | Data length byte | Return data high byte | Return data low byte | Return data (N+1) high byte | Return data (N+1) low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|------------------|-----------------------|----------------------|-----------------------------|----------------------------|-------------------------|--------------------------|
| 0X01    | 0X03          |                  |                       |                      |                             |                            |                         |                          |

Example 1: Read output voltage value (0x000D)

PC sends: 01 03 00 0D 00 02 55 C8

(Starting address is 0x000D, length is 2 registers, CRC16 checksum result is 0xC855)

Power supply returns: 01 03 04 40 A0 00 00 EF D1

(0x40A00000 is 5.000V, 0xD1EF is the CRC16 checksum result)

Example 2: Read output voltage and current values (0x000D ~ 0x0010)

PC sends: 01 03 00 0D 00 04 D5 CA

(Starting address is 0x000D, length is 4 registers, CRC16 checksum result is 0xCAD5)

PC sends: 01 03 08 40 A0 00 00 40 00 00 00 24 2D

(0x40A00000 represents 5.000V, 0x40000000 represents 2.0000A, and 0x2D24 is the CRC16 checksum result)

## Function Code 0X06

PC sends: 8 Byte

| Address | Function code | Write address high byte | Write address low byte | Write data high byte | Write data low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|-------------------------|------------------------|----------------------|---------------------|-------------------------|--------------------------|
| 0X01    | 0X06          |                         |                        |                      |                     |                         |                          |

Power supply returns: 8 Byte

| Address | Function code | Write address high byte | Write address low byte | Write data high byte | Write data low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|-------------------------|------------------------|----------------------|---------------------|-------------------------|--------------------------|
| 0X01    | 0X06          |                         |                        |                      |                     |                         |                          |

Note: Function code 06 is not supported for float-type parameter registers. Function code 06 only supports single register operations.

Example 1: Turn on power output (0x000B)

PC sends: 01 06 00 0B 00 01 39 C8 (Write address is 0x000B, write data is 0x0001, CRC16 checksum result is 0xC839)

Power supply returns: 01 06 00 0B 00 01 39 C8 (Write address is 0x000B, write data is 0x0001, CRC16 checksum result is 0xC839)

## Function Code 0X10

PC send: 9+N\*2 Byte

| Address | Function code | Starting address high byte | Starting address low byte | Number of registers high byte | Number of registers low byte | Data length byte | Write data high byte | Write data low byte | Write data (N+1) high byte | Write data (N+1) low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|----------------------------|---------------------------|-------------------------------|------------------------------|------------------|----------------------|---------------------|----------------------------|---------------------------|-------------------------|--------------------------|
| 0X01    | 0X10          |                            |                           |                               |                              |                  |                      |                     |                            |                           |                         |                          |

Power supply returns: 8 Byte

| Address | Function code | Starting address high byte | Starting address low byte | Data length high byte | Data length low byte | CRC16 checksum low byte | CRC16 checksum high byte |
|---------|---------------|----------------------------|---------------------------|-----------------------|----------------------|-------------------------|--------------------------|
| 0X01    | 0X10          |                            |                           |                       |                      |                         |                          |

Example 1: Set voltage and current values simultaneously (0x0001 ~ 0x0004)

PC send: 01 10 00 01 00 04 08 40 A0 00 00 40 00 00 00 FA 43

(Starting address is 0x0001, length is 4 registers, 0x40A00000 is 5.000V, 0x40000000 is 2.0000A, CRC16 checksum result is 0x43FA)

Power supply returns: 01 10 00 01 00 04 90 0A (Starting address is 0x0001, 4 registers, CRC16 checksum result is 0x0A90)